

Sec: XI_STU_IC_IIT

Time: 1.5 Hrs.

WARM UP – FTM-3

Date:01-07-2024

Max.Marks:180

JEE MAIN Model

PHYSICS

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 1 – 10)	Questions with Single Correct Choice	4	-1	10	40
Sec-II (Q.N : 11 – 15)	Questions with Integer Answer Type(+/-Number)	4	-1	5	20
Total				15	60

CHEMISTRY

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 16 – 25)	Questions with Single Correct Choice	4	-1	10	40
Sec-II (Q.N : 26 – 30)	Questions with Integer Answer Type (+/- Number)	4	-1	5	20
Total				15	60

MATHEMATICS

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec-I (Q.N : 31 – 40)	Questions with Single Correct Choice	4	-1	10	40
Sec-II (Q.N : 41 – 45)	Questions with Integer Answer Type (+/- Number)	4	-1	5	20
Total				15	60

Exam Syllabus

PHYSICS	40% : Motion under gravity, Graphs 40% : KINEMATICS: Uniformly accelerated motion (1D) 20% : MATHEMATICAL PHYSICS: Application of differentiation in physics, Maxima & Minima Integration and its application in physics Unit and Dimension: Introduction of physics, physical quantities and physical laws Fundamental and derived quantities and ways to write dimension, Unit and Dimension Uses of dimensional analysis (Principle of homogeneity) Unit and Dimension: Significant figures, Errors in measurements Problems discussion from material KINEMATICS: Frame of Reference, Mathematical Expression and definition of position, distance and displacement Avg velocity & instantaneous velocity, Avg speed & instantaneous speed, acceleration
CHEMISTRY	40% : Equivalent weight and related numericals (excluding redox) Normality and equivalent concept, 40% : Percentage calculation of elements, Formula representation of molecule: (i) Empirical formula, (ii) Molecular formula, (iii) Formula unit-factor of acid, base and salts 20% : Nature of Matter, Properties of matter & their measurement SI unit, Mass weight, Volume temperature, Need for standard reference, Scientific notation, Precision & accuracy, Significant figure and calculation involving significant figure. Law of chemical combination : (i) Law of conservation of mass, (ii) Law of constant combination, (iii) Law of multiple proportion, (v) Gay Lussac's law, (vi) Avagadro law, (vii) Dalton's atomic theory Atomic mass Mole concept: Calculation based on mole concept, concept of gram atom, gram molecule Percentage by mass (w/w), percentage by volume (v/v) Concentration terms: Molarity, molality, mole fraction and their interconversions, relation between density & molarity Concentration terms and their interconversions, relation between density & molarity Question practice on concentration Concept of limiting reagent : Use of stoichiometry in balanced equation & limiting reagent concept
MATHEMATICS	40% : maximum and minimum value of trigonometric expressions, analysis of the form $y = a \sin x + b \cos x$. conditional identities, sum of Trigonometric series. 40% : Domain and ranges of trigonometric function, Graph of trigonometric functions. Trigonometric ratio of multiple angles, sub-multiple angles. 20% : SEQUENCE AND SERIES Introduction of Sequence and Series, nth term of AP. Properties of A.P. Sum of n term of an AP, Arithmetic Mean, nth term of G.P., sum of n term of GP. Properties of G.P., Geometric mean, A.G.P., Introduction to HP, HM, Relation between AM, GM and HM Difference method, Vn method, special series, TRIGONOMETRIC FUNCTIONS Angle, system of measurement of angles, trigonometric function, Values of trigonometric function for some specific angles, Trigonometric ratios of allied angles, Transforming product into sum or difference, Transforming the sum or difference into product.

Rough Work

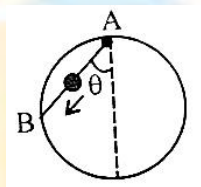
(SINGLE CORRECT ANSWER TYPE)

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

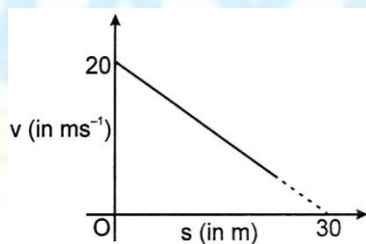
1. A particle moves a distance x in time t according to equation $x = (t+1)^{-1}$. The acceleration of particle is proportional to
A) (velocity)^{3/2} B) (distance)²
C) (distance)⁻² D) (velocity)^{2/3}
2. A body of mass 10 kg is moving with a constant velocity of 10 m/s. When a constant force acts for 4 seconds on it, it moves with a velocity 2 m/sec in the opposite direction. The acceleration produced in it is:
A) 3 m/sec² B) - 3 m/sec²
C) 0.3 m/sec² D) - 0.3 m/sec²
3. A motor car moving with a uniform speed of 20 m/sec comes to stop on the application of brakes after travelling a distance of 10 m. Its acceleration is:
A) 20 m/sec² B) - 20 m/sec²
C) - 40 m/sec² D) + 2 m/sec²
4. A particle moves in a straight line with a constant acceleration. It changes its velocity from 10 m/s to 20 m/s while passing through distance 135 m in t second. The value of t is
A) 12 B) 9
C) 10 D) 1.8
5. An object accelerates from rest to a velocity 27.5 m/s in 10 sec then find distance covered by object in next 10 sec:
A) 550 m B) 137.5 m
C) 412.5 m D) 275 m
6. A body starts from rest and is uniformly accelerated for 30 s. The distance travelled in the first 10 s is x_1 , next 10 s is x_2 and the last 10 s is x_3 . Then $x_1 : x_2 : x_3$ is the same as
A) 1 : 2 : 4 B) 1 : 2 : 5
C) 1 : 3 : 5 D) 1 : 3 : 9

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7. A ball is dropped from a height of 20 m and rebounds with a velocity which is $\frac{3}{4}$ of the velocity with which it hits the ground. What is the time interval between the first and second bounces
 A) 3 sec. B) 4 sec.
 C) 5 sec. D) 6 sec.
8. A body is projected vertically upwards. It reaches the maximum vertical height h in time t . In $t/2$, the height covered is
 A) $h/2$ B) $2h/5$
 C) $3h/4$ D) $h/4$
9. A bead is free to slide down a smooth wire tightly stretched between points A and B on a vertical circle. If the bead starts from rest of A, the highest point on the circle.



- A) its velocity v on arriving at B is proportional to $\sin\theta$
 - B) its velocity v on arriving at B is proportional to $\tan\theta$
 - C) time to arrive to B is proportional to $\cos\theta$
 - D) time to arrive to B is independent of θ
10. If the velocity v of a particle moving along a straight line decreases linearly with its displacement s from 20 ms^{-1} to a value approaching zero at $s = 30 \text{ m}$, the acceleration of the particle at $s = 15 \text{ m}$ is



- A) $\frac{2}{3} \text{ ms}^{-2}$ B) $-\frac{2}{3} \text{ ms}^{-2}$
- C) $\frac{20}{3} \text{ ms}^{-2}$ D) $-\frac{20}{3} \text{ ms}^{-2}$

Rough Work

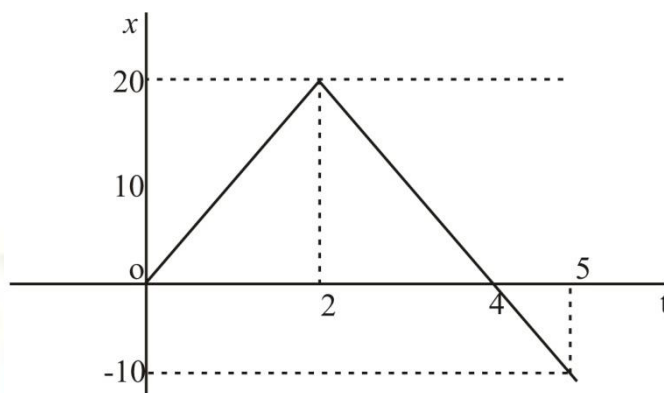
SECTION-II

(Integer Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer is in the decimals. Question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

11. The diagram shows the displacement-time graph for a particle moving in a straight line. The average speed for the interval $t = 0$ to $t = 5$ is V . Then value of $\frac{V}{5}$



12. A particle moves along a straight line and its position depends on time as $x = 4t - t^2$. Then for first 5 sec. magnitude average velocity.....
13. The heat generated in a circuit is given by $Q = I^2 R t$, where I is current, R is resistance, and t is time. If the percentage errors in measuring I , R and t are 2%, 1% and 1%, respectively, then the maximum error in measuring heat will be.....
14. A ball is projected vertically upward with an initial velocity of 50 ms^{-1} at $t = 0\text{s}$. At $t = 2\text{s}$. another ball is projected vertically upward with same velocity. At $t = \underline{\hspace{2cm}}$ s, second ball will meet the first ball ($g = 10 \text{ ms}^{-2}$)
15. If $y = 5x^4 + 5x + 6$, then $\frac{dy}{dx}$ at $x = 0$

Rough Work

CHEMISTRY

Max.Marks:60

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

16. Match the column I with column II.

Column-I		Column-II	
(a)	88 g of CO ₂	(i)	0.25 mol
(b)	6.022×10^{23} molecules of H ₂ O	(ii)	2 mol
(c)	5.6 litres of O ₂ at STP	(iii)	1 mol
(d)	96 g of O ₂	(iv)	6.022×10^{23} molecules
(e)	1 mol of any gas	(v)	3 mol

A) (a) – iii, (b) – ii, (c)-v, (d)-iv, (e)-i

B) (a) – i, (b) – ii, (c)-iii, (d)-iv, (e)-v

C) (a) – ii, (b) – iii, (c)-i, (d)- v, (e)-iv

D) (a) – v, (b) – iv, (c)-ii, (d)-iii, (e)-i

17. Calculate the molarity of NaOH in the solution prepared by dissolving its 4g in enough water to form 250 mL of the solution.

A) 0.2 M

B) 0.3 M

C) 0.4 M

D) 0.5 M

18. The complete combustion of 0.492 g of an organic compound containing 'C', 'H' and 'O' gives 0.793g of CO₂ and 0.442 g of H₂O. The percentage of oxygen composition in the organic compound is

A) 46.1

B) 43.2

C) 51.2

D) 39.6

19. Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/ml. The molarity of the solution is

A) 1.78 M

B) 2.00 M

C) 2.05 M

D) 2.22 M

Rough Work

SECTION-II

(Integer Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer is in the decimals. Question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. Given that the abundances of isotopes ^{54}Fe , ^{56}Fe and ^{57}Fe are 5%, 90% and 5%, respectively, the atomic mass of Fe is..... (In nearest integer)
27. The total number of electrons in one molecule of urea.....
28. Sum of the n-factor of following -
Salt (i) Potash Alum
(ii) Potassium chloride
29. An organic compound gives 0.220 g of CO_2 and 0.126 g of H_2O on complete combustion. If the % of carbon is 24 then the % hydrogen is $\times 10^{-1}$. (Nearest integer)
30. Two elements A and B which form 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A istimes of atomic weight of B.

Rough Work

MATHEMATICS**Max.Marks:60****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. If $\sin^2 10^\circ \cdot \sin 20^\circ \sin 40^\circ \sin 50^\circ \sin 70^\circ = \alpha - \frac{1}{16} \sin 10^\circ$ then $\frac{16 + \alpha^{-1}}{10}$ is equal to:

- A) 8
B) 10
C) 2
D) None

32. The value of $\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \dots \cos \frac{\pi}{2^{10}} \cdot \sin \frac{\pi}{2^{10}}$ is

- A) $\frac{1}{512}$
B) $\frac{1}{1024}$
C) $\frac{1}{256}$
D) $\frac{1}{2}$

33. The maximum value of $(3 \sin \theta - 1)^2 - 2$ is:

- A) 2
B) 12
C) 14
D) 16

34. $\sec^2 \theta$ lies in the interval:

- A) $(0, \infty)$
B) $(-\infty, -1] \cup [1, \infty)$
C) $R - (-\infty, 1)$
D) $(1, \infty)$

35. If $A + B = \frac{\pi}{4}$ then $\tan A + \tan B + \tan A \cdot \tan B$ equals

- A) 1
B) 2
C) 3
D) None

Rough Work

36. Let G be geometric mean of two positive numbers a and b and M be arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$. If $\frac{1}{M} : G$ is $4 : 5$, then $a : b$ can be:
- A) $1 : 4$ B) $1 : 2$
C) $2 : 3$ D) $3 : 4$
37. For any three positive real numbers a, b and c
 $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then
- A) a, b, c are in G.P. B) b, c, a are in G.P.
C) b, c, a are in A.P. D) a, b, c are in A.P.
38. If m and M are minimum and maximum values of $4 + \frac{1}{2} \sin^2 2x - 2 \cos^4 x, x \in R$ then $M - m$ is equal to:
- A) $\frac{9}{4}$ B) $\frac{15}{4}$
C) $\frac{7}{4}$ D) $\frac{1}{4}$
39. The value of $\frac{2 \tan \theta}{1 + \tan^2 \theta}$ at $\theta = 7\frac{1}{2}^\circ$ is:
- A) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ B) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
C) $\frac{\sqrt{5}-1}{4}$ D) $\frac{\sqrt{5}+1}{4}$
40. If A, B, C are angles of a triangle then $\sin 2A + \sin 2B + \sin 2C$ equals:
- A) $4 \cos A \cos B \cos C$ B) $4 \sin A \sin B \sin C$
C) $1 + 4 \cos A \cos B \cos C$ D) $-1 - 4 \sin A \sin B \sin C$

Rough Work

SECTION-II

(Integer Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer is in the decimals. Question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

41. The maximum value of $\sqrt{6} \sin \alpha + \sqrt{3} \cos \beta + 3$ is _____, where ' α ' & ' β ' are the roots of a quadratic equation whose discriminant is zero.
42. The value of $\sin \theta (3 - 4 \sin^2 \theta)$ at $\theta = 15^\circ$ is $\frac{1}{\alpha}$ then α^2 equals _____.
43. The maximum value of ' n ', $n \in N$ for which $\sin \theta = -\frac{1}{2}$ has four solutions in the interval $[0, n\pi]$ is _____.
44. If T_r denotes the r^{th} term of an A.P. and $T_m + T_{2n+m} = kT_{m+n}$ where $k \in N$ then k equals _____.
45. The value of $\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \dots + \sin \frac{12\pi}{7}$ is ' k ' then $2k^2$ equals _____.

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Rough Work